

Train_Test_split

October 31, 2025

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[1]: # =====  
# Import libraries  
# =====  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
from scipy import stats  
from sklearn.model_selection import train_test_split  
  
sns.set(style="whitegrid", palette="muted", font_scale=1.1)  
plt.rcParams["figure.figsize"] = (6, 4)
```

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[2]: # =====  
# Load dataset  
# =====  
df = pd.read_csv("chla_dataset.csv")  
  
print("Chl-a dataset loaded successfully!\n")  
display(df.head())
```

Chl-a dataset loaded successfully!

	temp	salt	chla
0	27.506178	34.005938	8.006679
1	25.783341	34.858689	5.883912
2	28.117596	33.605527	8.108461
3	24.122508	34.319593	5.246205
4	25.288687	33.493397	4.106262

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[3]: # =====  
# Exploratory Data Analysis (EDA)  
# =====  
  
# ---- Summary statistics ----  
print("\n Summary statistics:")  
display(df.describe())
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# ---- Distribution plots ----
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
for ax, col in zip(axes, df.columns):
    sns.histplot(df[col], kde=True, ax=ax)
    ax.set_title(f"Distribution of {col}")
plt.tight_layout()
plt.show()

# ---- Scatter plots & correlations ----
corr_temp = df['chla'].corr(df['temp'])
corr_salt = df['chla'].corr(df['salt'])

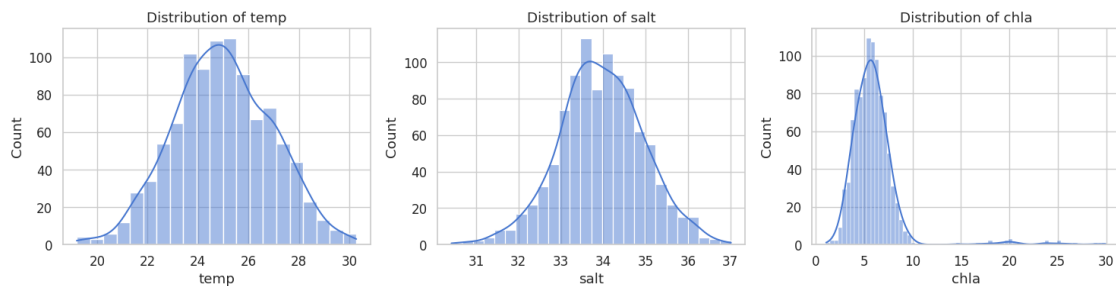
sns.lmplot(data=df, x='temp', y='chla', line_kws={'color':'red'})
plt.title(f"Chl-a vs Temp (r = {corr_temp:.2f})")
plt.show()

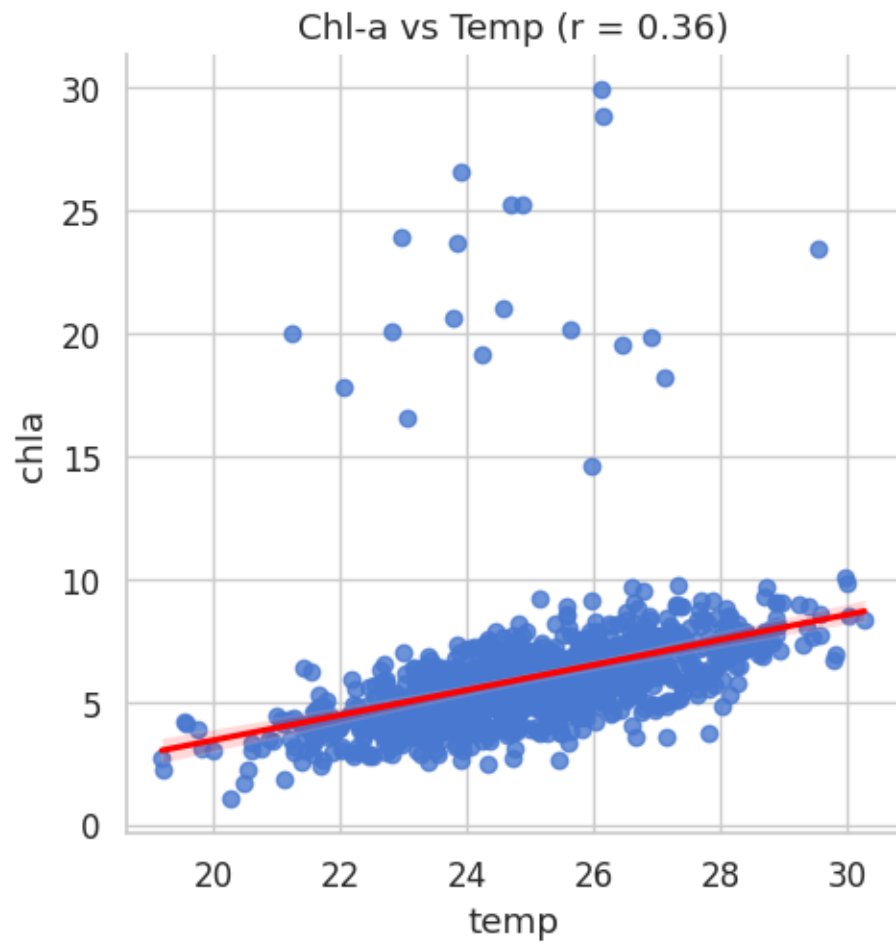
sns.lmplot(data=df, x='salt', y='chla', line_kws={'color':'red'})
plt.title(f"Chl-a vs Salt (r = {corr_salt:.2f})")
plt.show()

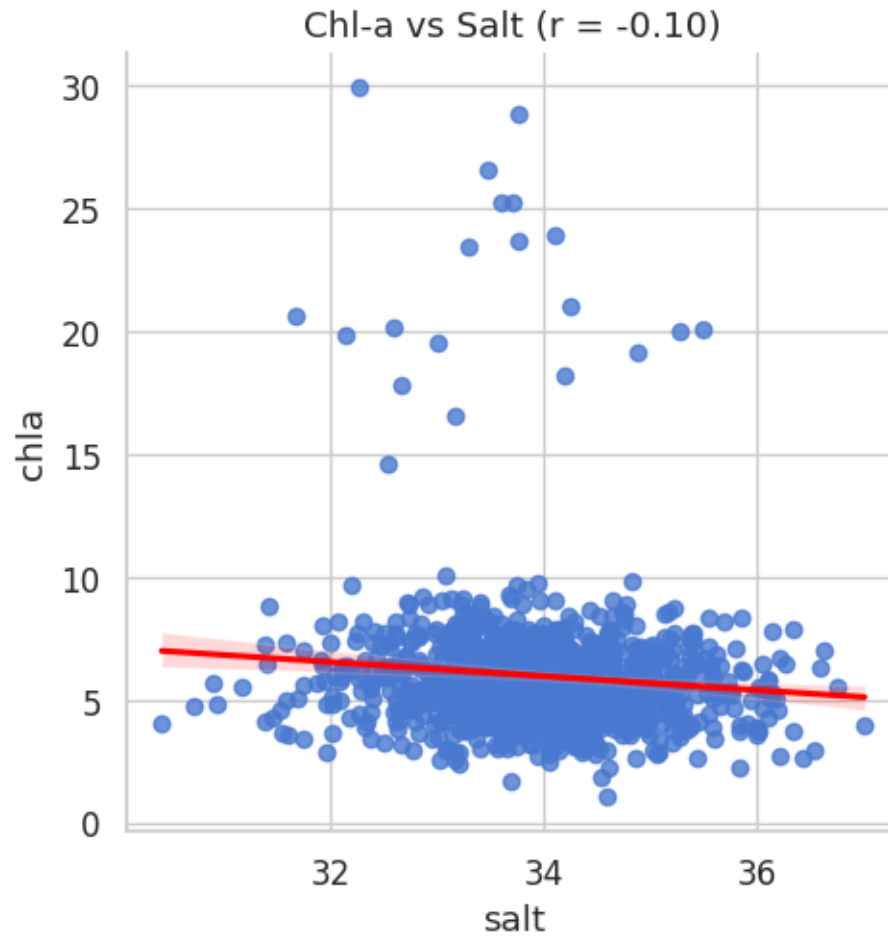
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Summary statistics:

	temp	salt	chla
count	1000.000000	1000.000000	1000.000000
mean	24.982510	33.953689	5.988593
std	1.933228	0.994764	2.744487
min	19.191071	30.417937	1.085851
25%	23.683816	33.314328	4.625095
50%	24.909789	33.947444	5.700538
75%	26.360994	34.614775	6.736032
max	30.275390	37.014712	29.956012







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[4]: # =====
# Remove outliers (Z-score method)
# =====
def remove_outliers(df):
    z = np.abs(stats.zscore(df))
    return df[(z <= 3).all(axis=1)]

df_clean = remove_outliers(df)

print(f"\n Outlier removal complete: {len(df)} → {len(df_clean)} rows")
```

Outlier removal complete: 1000 → 975 rows

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[5]: # =====
# Train-Test Split
# =====
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X = df_clean[['temp', 'salt']]
y = df_clean['chla']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("\n Train-Test Split Summary:")
print(f"X_train: {X_train.shape}")
print(f"X_test: {X_test.shape}")
print(f"y_train: {y_train.shape}")
print(f"y_test: {y_test.shape}")

```

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Train-Test Split Summary:
X_train: (780, 2)
X_test: (195, 2)
y_train: (780,)
y_test: (195,)

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[6]: X_train.to_csv("X_train.csv", index=False)
      X_test.to_csv("X_test.csv", index=False)
      y_train.to_csv("y_train.csv", index=False)
      y_test.to_csv("y_test.csv", index=False)

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